



The formula with irrational roots

HANDOUT

GOAL

In this lesson

Solve quadratic equations whose roots are irrational, and write the roots exactly with $\sqrt{D}$ simplified.

BY THE END OF THE LESSON

- ☐ I can solve a quadratic equation whose roots are irrational: simplify $\sqrt{D}$ and write the exact roots.

1 class period · about 45 min

WHAT YOU NEED FROM BEFORE

- ↔ Lesson 1.3 · The quadratic formula
↔ Square roots

HOOK

The second garden

Lesson 1.3 left the second garden unsolved. It is 3 m longer than it is wide, and its area is 30 m². Its width x satisfies $x^2 + 3x - 30 = 0$.

Level 1 · threshold

Work out the discriminant of $x^2 + 3x - 30 = 0$.

The formula still works when D is not a perfect square. The only question is how to write the answer.

EXPLAIN

Exact roots

An exact answer, then a rounded one

For the garden, $a = 1$, $b = 3$ and $D = 129$:

$$x_{1,2} = \frac{-3 \pm \sqrt{129}}{2}.$$

These are the **exact** roots. $\sqrt{129}$ cannot be simplified: $129 = 3 \cdot 43$ has no perfect-square factor.

A width is positive, so the width is $x_2 = \frac{-3 + \sqrt{129}}{2}$ m. The other root is negative. This exact value is the answer. For a garden it is useful to know the width roughly too, so we also round it, **at the very end**. $\sqrt{129} \approx 11.358$, so, to 0.01 m,

$$x_2 \approx \frac{-3 + 11.358}{2} \approx 4.18 \text{ m.}$$

Check: $4.18 \cdot 7.18 \approx 30.01$. That is close to 30, as a rounded answer should be.

Keep the root exact until the last step. Round only when the question asks for a decimal answer, or when a decimal value is useful, as for the garden.

Simplify the root of D

Solve $x^2 - 6x + 7 = 0$. Here $a = 1$, $b = -6$, $c = 7$.

$$D = (-6)^2 - 4 \cdot 1 \cdot 7 = 36 - 28 = 8.$$

8 is not a perfect square, but it has the perfect-square factor 4: $\sqrt{8} = \sqrt{4 \cdot 2} = 2\sqrt{2}$. So

$$x_{1,2} = \frac{6 \pm 2\sqrt{2}}{2}.$$

Divide the whole numerator

The fraction bar divides the **whole** numerator by 2, so **each** term is divided:

$$\frac{6 \pm 2\sqrt{2}}{2} = \frac{6}{2} \pm \frac{2\sqrt{2}}{2} = 3 \pm \sqrt{2}.$$

Another way to see it: take out the common factor, $\frac{2(3 \pm \sqrt{2})}{2} = 3 \pm \sqrt{2}$.

A common mistake is $3 \pm 2\sqrt{2}$: only the 6 was divided. With plain numbers the mistake is easy to see:

$$\frac{6+4}{2} = 5 = 3+2, \text{ not } 3+4.$$

Lesson 1.2's method gives the same roots: $x^2 - 6x + 7 = 0$ becomes $(x-3)^2 = 2$, so $x = 3 \pm \sqrt{2}$.

When is it simple enough?

Roots like these are in their simplest form when

1. no perfect-square factor is left under the root ($\sqrt{8}$ becomes $2\sqrt{2}$), and
2. no whole number greater than 1 divides all three: the term without the root, the number in front of the root, and the denominator.

For example, in $\frac{6 \pm 2\sqrt{3}}{4}$ the numbers 6, 2 and 4 are all divisible by 2:

$$\frac{6 \pm 2\sqrt{3}}{4} = \frac{3 \pm \sqrt{3}}{2}.$$

But $\frac{-3 \pm \sqrt{129}}{2}$ stays as it is: 3, 1 and 2 have no common factor.

Both ways of writing the roots are fine: $x = 3 \pm \sqrt{2}$, or $x_1 = 3 - \sqrt{2}$ and $x_2 = 3 + \sqrt{2}$. In the answer boxes on this site, type $\sqrt{}$ as sqrt and $\pm$ as +/-, e.g. 3 +/- sqrt(2).

Worked examples

Example 1 • $x^2 - 4x - 1 = 0$

COEFFICIENTS

$$a = 1, b = -4, c = -1.$$

DISCRIMINANT

$$D = (-4)^2 - 4 \cdot 1 \cdot (-1) = 16 + 4 = 20, \text{ and } \sqrt{20} = \sqrt{4 \cdot 5} = 2\sqrt{5}.$$

FORMULA

$$x_{1,2} = \frac{4 \pm 2\sqrt{5}}{2}.$$

SIMPLIFY

Divide both terms by 2: $x_{1,2} = 2 \pm \sqrt{5}$, that is, $x_1 = 2 - \sqrt{5}$ and $x_2 = 2 + \sqrt{5}$.

Example 2 • When $a \neq 1$: $2x^2 - 6x + 3 = 0$

COEFFICIENTS

$$a = 2, b = -6, c = 3.$$

DISCRIMINANT

$$D = (-6)^2 - 4 \cdot 2 \cdot 3 = 36 - 24 = 12, \text{ and } \sqrt{12} = 2\sqrt{3}.$$

FORMULA

$$x_{1,2} = \frac{6 \pm 2\sqrt{3}}{2 \cdot 2} = \frac{6 \pm 2\sqrt{3}}{4}.$$

SIMPLIFY

6, 2 and 4 have the common factor 2 (not 4): $x_{1,2} = \frac{3 \pm \sqrt{3}}{2}.$

ROUND, IF ASKED

$\sqrt{3} \approx 1.732$, so $x_1 \approx 0.63$ and $x_2 \approx 2.37$.

SUMMARY

What to remember

- When the coefficients are whole numbers and $D > 0$ is not a perfect square, the roots are irrational. Leave $\sqrt{D}$ as a root: the roots are then **exact**.
- **Simplify $\sqrt{D}$** : find the largest perfect-square factor and take out its root, e.g. $\sqrt{20} = \sqrt{4 \cdot 5} = 2\sqrt{5}$.
- **Divide the whole numerator**: $\frac{4 \pm 2\sqrt{5}}{2} = 2 \pm \sqrt{5}$, not $2 \pm 2\sqrt{5}$. Cancel the fraction only by a common factor of all three numbers: $\frac{6 \pm 2\sqrt{3}}{4} = \frac{3 \pm \sqrt{3}}{2}$.
- **Round at the very end**, and only when the question asks for a decimal answer or a decimal value is useful.

NEXT LESSON

So far D has always been positive. What does the formula give when $D = 0$, or when D is negative, and can we tell the number of roots before solving?

Lesson 1.5 · The discriminant: how many roots?